

Caching strategies

To maintain speed, SSDs employ caches. Many drives include a DRAM cache, a small memory chip that stores mapping tables and buffers data. DRAM improves random performance and reduces write amplification but adds cost. Budget drives omit DRAM and instead borrow a little system RAM via the Host Memory Buffer (HMB) feature; this helps but can't match on-board DRAM. Performance drops are most noticeable when multitasking, so heavy users should prioritise DRAM.

SSDs also use SLC caches, regions of NAND treated as single-level cells to accelerate writes. Static caches reserve a fixed area, delivering consistent burst speed but falling off sharply when full. Dynamic caches expand when there is lots of free space, offering longer bursts at first but shrinking as the drive fills, leading to unpredictable sustained speeds. Low-cost drives may forgo SLC caching entirely and write directly to TLC or QLC, reducing cost but also performance. If you regularly transfer large files, choose a drive with a generous static cache; if you seldom write huge files, a dynamic cache is fine.

Picking the right NVMe drive

Capacity and configuration. In 2025, AAA games can exceed 150 GB and 4K video projects occupy hundreds of gigabytes. A 2 TB



NVMe drive provides comfortable room for gaming, editing and productivity. Smaller drives cost less but fill quickly and may have lower endurance because they use fewer NAND chips. High-capacity drives (4–8 TB) are best for media libraries and AI datasets but cost more and may run hotter. Many builders choose a dual-drive setup: a fast NVMe SSD for the operating system and active projects and a larger, slower drive or network storage for archives.

EXPLAINED: PCIE GEN 4 VS GEN 5 NVME SSDS			
Feature	PCIe Gen 4 SSD	PCIe Gen 5 SSD	What It Means for You
Launch Window	2019 – present	2023 – present	Gen 4 remains mainstream; Gen 5 still early-adopter tech.
Maximum Interface Bandwidth (x4)	~8 GB/s (7,880 MB/s theoretical)	~16 GB/s (15,750 MB/s theoretical)	Gen 5 doubles potential throughput.
Typical Sequential Read Speeds	5,000 – 7,400 MB/s	10,000 – 14,000 MB/s	Real-world gains appear mainly in large-file transfers.
Typical Sequential Write Speeds	4,000 – 6,500 MB/s	8,000 – 12,000 MB/s	Gen 5 excels in sustained, high-volume writes.
Random 4K Performance (IOPS)	~1 M IOPS	~1.4 M IOPS +	Better queue-depth response and multi-task performance.
Controller Heat Output	Moderate	Very High	Gen 5 drives often ship with active heatsinks or fans.
Ideal Use Case	Gaming, everyday productivity, 4K editing	8K video, AI workloads, professional data handling	Choose Gen 5 only if your workflow truly needs it.
Compatibility	Works on PCIe Gen 4 and Gen 5 slots	Backward-compatible but throttles to Gen 4 speed	Check motherboard support before buying.
Price per GB (2025 India)	₹6 – ₹9 / GB	₹9 – ₹15 / GB	Gen 5 carries a noticeable premium.
Example Models	Samsung 990 Pro, WD Black SN850X, Crucial T500	Seagate FireCuda 540, ADATA XPG Gammix S70 Blade Gen 5	Illustrative only – check regional SKUs.